ARDUPILOT NAVIGATION GUIDE: GPS JAMMED, SPOOFED & NO-GPS ENVIRONMENTS

Flight Dynamics, EKF3 Failure Cascades & Safe Parameter Configuration Matrix

Applicable Fleet: All ArduPilot Fixed-Wing, VTOL & Quadplane Platforms with EKF3 Core

1. Executive Summary & Threat Taxonomy

Operating unmanned air vehicles (UAVs) in contested, electronically degraded airspace requires an exact understanding of how ArduPilot's Extended Kalman Filter (EKF3) and Attitude & Heading Reference System (AHRS) interact with corrupted sensor inputs.

Threat Regime	Physical Manifestation	Autopilot Vulnerability	Primary Countermeasure
RF Jamming	Broadband carrier noise ($C/N_0 < 28$ dB-Hz). Receiver **stays powered** with jumping 3D fixes (±50–400 m/s).	**DCM Fallback Hunting:** Autopilot drops to legacy DCM, feeding raw noise to navigation; causes severe wandering.	Lock out DCM (`AHRS_OPTIONS = 3`) and inhibit timeout reset (`EK3_OPTIONS = 1`).
GPS Spoofing	Hostile transmitters broadcast coherent false ephemeris; injects coordinate step-jumps or deceptive drift vectors.	**Trajectory Hijacking:** Autopilot accepts false coordinates, diverging miles off course or into terrain.	Tighten horizontal gate (`EK3_GLITCH_RAD = 15`) and isolate attitude (`AHRS_GPS_USE = 0`).
Total Denial (Cutoff)	Complete loss of GNSS signal ($fix_type = 0$, $sats = 0$) due to antenna severance or hardware cutoff (`SIM_GPS1_ENABLE 0`).	**Wind Drift:** Position estimate diverges if horizontal aerodynamic wind is unknown or unmodelled.	Calibrate airframe drag (`EK3_DRAG_BCOEF_X/Y = 25`) and pitot True Airspeed sensor.
Hover Denial	GPS denied while VTOL or quadplane is stationary in hover ($V_{\text{air}} \approx 0$ m/s).	**Hover Departure:** Pitot tube has no dynamic pressure; cannot estimate aerodynamic wind vector.	Optical flow camera velocity fusion (`FLOW_ENABLE = 1`, `EK3_SRC1_VELXY = 5`).

2. Normal Jammed Scenario: Unconfigured vs. Configured Execution

A common misconception is that ArduPilot automatically switches off its GPS when jammed. **It does not.** The GPS module continues transmitting noisy coordinates. Understanding the internal failure sequence is critical:

Default / Unconfigured Autopilot (Failure Chain)	Properly Configured Autopilot (Safe Execution)
1. DCM Fallback Trap (`AHRS_OPTIONS = 0`): EKF3 flags innovation anomaly (`pos_horiz_ratio > 1.0`) and rejects GPS. However, `AP_AHRS` sees receiver status is still ≥ 3D FIX and automatically falls back to legacy DCM. DCM has **no dead reckoning filter**; it feeds raw jumping coordinates straight to L1 navigation, commanding erratic banks left/right (**hunting/wandering**).	**1. Innovation Gate Rejection (`EK3_GLITCH_RAD = 15`):** EKF3 checks incoming coordinates against inertial propagation. Corrupted jumps exceeding 15 m are rejected at the gate.
2. Attitude Horizon Corruption (`AHRS_GPS_USE = 1`): Centrifugal turn compensation calculates $a_{\text{cent}} = \omega_z \cdot V_{\text{gps}}$. Erratic GPS groundspeed spikes corrupt the artificial horizon, inducing violent wing-rocking.	**2. DCM Lockout Active (`AHRS_OPTIONS = 3`):** Fallback to DCM is strictly inhibited (Bit 0: Plane, Bit 1: VTOL). Navigation remains locked inside EKF3 Dead Reckoning.
3. Position Timeout Reset (`EK3_OPTIONS = 0`): After 5–10 s of rejected GPS, EKF3's position timeout expires. It resets its position state directly onto the bad GPS coordinates!	**3. Horizon Isolated & Reset Inhibited (`AHRS_GPS_USE 0`, `EK3_OPTIONS 1`):** Attitude uses pitot airspeed; horizon stays level. EKF3 never resets to bad GPS; it holds dead reckoning indefinitely.
Result: Aircraft weaves violently, loses altitude, or departs controlled flight.	**Result:** Smooth, straight waypoint tracking on internal aerodynamic Dead Reckoning. Zero hunting or wandering.

3. Safe Parameter Configuration by Threat Regime

Table 1: Electronic Warfare RF Jamming Countermeasures

Parameter	Default	Target	Threat Scope	Operational Function & Source Mechanism	Terminal Command
AHRS_OPTIONS	0	3	Jamming Hunting	**DCM Lockout:** Bit 0 (Plane) + Bit 1 (VTOL). Inhibits automatic fallback to legacy DCM when GPS reports degraded 3D fix. Eliminates wandering.	param set AHRS_OPTIONS 3
EK3_OPTIONS	0	1	Position Reset	**Jamming Expected:** Bit 0 (JammingExpected). Prevents EKF3 from resetting position to bad GPS upon timeout; locks EKF in Dead Reckoning.	param set EK3_OPTIONS 1
AHRS_GPS_USE	1	0	Wing-Rocking	**Attitude Isolation:** Disconnects GPS ground velocity from centrifugal turn compensation ($a_{\text{cent}} = \omega_z \cdot V_{\text{gps}}$). Keeps horizon true.	param set AHRS_GPS_USE 0

Table 2: Adversarial GPS Spoofing & Innovation Step Gating

Parameter	Default	Target	Threat Scope	Operational Function & Source Mechanism	Terminal Command
EK3_GLITCH_RAD	25.0 m	15.0 m	Trajectory Hijack	**Step-Jump Gate:** Maximum allowable horizontal coordinate jump between consecutive GPS updates before EKF rejects it as anomalous.	param set EK3_GLITCH_RAD 15
GPS_GNSS_MODE	0	65	Single-Band Spoof	**Multi-Constellation Diversity:** Forces receiver to track GPS + Galileo + GLONASS simultaneously, preventing single-band spoofing.	param set GPS_GNSS_MODE 65

Table 3: Total GPS Denial & Aerodynamic Dead Reckoning (Fixed-Wing Cruise)

Parameter	Default	Target	Threat Scope	Operational Function & Source Mechanism	Terminal Command
EK3_DRAG_BCOEF_X	0.0	25.0	Wind Drift	**Longitudinal Drag Ballistics:** Fuselage drag coefficient (kg/m^2). Allows EKF3 to estimate wind vector from accelerometer drag vs airspeed.	param set EK3_DRAG_BCOEF_X 25
EK3_DRAG_BCOEF_Y	0.0	25.0	Crosswind Drift	**Lateral Drag Ballistics:** Airframe side-slip drag. Computes crosswind crab angle (β) to hold track in gusts.	param set EK3_DRAG_BCOEF_Y 25
ARSPD_USE	1	1	Speed Sensor Loss	**Mandatory Airspeed:** Pitot true airspeed (V_{tas}) is required to calculate $\vec{V}_{\text{ground}} = \vec{V}_{\text{air}} + \vec{W}_{\text{wind}}$.	param set ARSPD_USE 1
COMPASS_USE	1	1	Heading Drift	**Compass Fusion:** Accurately calibrated magnetometer is required to project true airspeed onto geographic North/East.	param set COMPASS_USE 1

Table 4: VTOL & Zero-Airspeed Hover Optical Flow Fallback

Parameter	Default	Target	Threat Scope	Operational Function & Source Mechanism	Terminal Command
SIM_FLOW_ENABLE	0	1	Hover Drift	**Camera Activation:** Enables downward optical flow camera sensor driver in SITL / hardware.	param set SIM_FLOW_ENABLE 1
EK3_FLOW_USE	1	1	Velocity Loss	**EKF Flow Fusion:** Directs EKF3 to ingest pixel displacement rates (rad/s) to observe horizontal ground speed.	param set EK3_FLOW_USE 1
EK3_SRC1_VELXY	3	5	Horizontal Wander	**Velocity Source Selection:** Switches primary horizontal velocity aiding from GPS (3) to Optical Flow (5).	param set EK3_SRC1_VELXY 5

4. Master Parameter Quick-Reference Table

This unified master matrix provides flight operators with an immediate single-page lookup for all contingency parameters:

#	Parameter	Default	Safe Target	Threat Scope	Single-Line Console Command
1	AHRS_OPTIONS	0	3	RF Jamming	`param set AHRS_OPTIONS 3`
2	EK3_OPTIONS	0	1	RF Jamming	`param set EK3_OPTIONS 1`
3	AHRS_GPS_USE	1	0	Jamming / Spoofing	`param set AHRS_GPS_USE 0`
4	EK3_GLITCH_RAD	25.0	15.0	Spoofing Gate	`param set EK3_GLITCH_RAD 15`
5	EK3_DRAG_BCOEF_X	0.0	25.0	No-GPS Cruise	`param set EK3_DRAG_BCOEF_X 25`
6	EK3_DRAG_BCOEF_Y	0.0	25.0	No-GPS Cruise	`param set EK3_DRAG_BCOEF_Y 25`
7	ARSPD_USE	1	1	No-GPS Cruise	`param set ARSPD_USE 1`
8	SIM_FLOW_ENABLE	0	1	No-GPS Hover	`param set SIM_FLOW_ENABLE 1`
9	EK3_SRC1_VELXY	3	5	No-GPS Hover	`param set EK3_SRC1_VELXY 5`

ONE-SHOT ANTI-JAM UPLOAD BLOCK (MAVProxy / SITL Console)

Copy and paste this script directly into your telemetry console before flight into contested airspace:

```
param set AHRS_OPTIONS 3
param set EK3_OPTIONS 1
param set AHRS_GPS_USE 0
param set EK3_GLITCH_RAD 15
param set EK3_DRAG_BCOEF_X 25
param set EK3_DRAG_BCOEF_Y 25
param set ARSPD_USE 1
```

RECOVERY SCRIPT (Upon Exiting Contested Airspace into Clear GPS)

When telemetry shows stable satellite constellation and continuous GPS lock ($fix_type \geq 3$):

```
param set AHRS_GPS_USE 1                    # Re-enable GPS velocity aiding for attitude horizon
param set SIM_GPS1_JAM 0                    # Clear simulated jamming noise (if in SITL)
param set SIM_GPS1_ENABLE 1                 # Re-enable simulated GPS receiver power (if in SITL)
```

5. Standard Operating Procedure (SOP) Flight Protocol

THE GOLDEN RULE OF DEAD RECKONING: THE WIND-LEARNING ORBIT

Dead Reckoning computes ground track from the vector triangle: $\vec{V}_{\text{ground}} = \vec{V}_{\text{air}} + \vec{W}_{\text{wind}}$.
If an aircraft flies into a jamming zone immediately after takeoff without learning the wind, EKF3 will assume zero wind and drift with the prevailing wind.

MANDATORY SOP: Climb to cruise altitude in good GPS and fly in **AUTO turns** or a **LOITER circle for 20–30 seconds**. This enables EKF3 to observe pitot true airspeed against groundspeed through 360° and converge the wind vector ($\vec{W}$) to < 0.5 m/s accuracy prior to ingress.

Flight Phase	Action Items & Operator Checks	Telemetry Annunciation & Expected Flight Dynamics
Phase 1: Pre-Flight	1. Perform pitot tube blow-check (±1.0 m/s). 2. Verify compass calibration quality ($fit_error < 2.0$). 3. Upload One-Shot Anti-Jam Parameter Block.	`ARSPD_ENABLE = 1`, `ARSPD_USE = 1`. MCDU / Visualizer displays `JAMMING EXPECTED` and `DCM LOCKOUT`.
Phase 2: Wind Orbit	Climb to 50–100 m AGL. Fly in `LOITER` for 20–30 s.	EKF3 converges horizontal wind states ($\vec{W}$). Telemetry displays stable wind speed and direction. Aircraft ready for ingress.
Phase 3: Jamming Entry	Enter contested EW zone. Jamming injected.	`STATUSTEXT: GPS_Glitch: pos step` or `pos timeout`. Aircraft maintains wings-level attitude, tracking waypoints smoothly. Position uncertainty ring expands at ≈ 1.2 m/s.
Phase 4: Recovery	Exit EW zone. GPS signals re-stabilize for 3–5 s.	EKF3 fuses GPS automatically. `pos_horiz_abs` transitions to `VALID`. Position uncertainty ring collapses (< 1.5 m). Restore: `param set AHRS_GPS_USE 1`.

EMERGENCY IMMEDIATE REACTION CHECKS (If Aircraft Hunts / Wanders in Contested Airspace):

1. **Verify DCM Lockout: Immediately execute `param set AHRS_OPTIONS 3`. (Stops DCM fallback).**
2. **Cut Degraded GPS Input: If coordinate jumping persists, execute `param set SIM_GPS1_ENABLE 0` (in SITL) or `param set GPS_TYPE 0`.**
3. **Confirm Airspeed Fusion: Check HUD/CDU for valid airspeed (> 12 m/s). Dead reckoning requires forward speed.**